MONTHLY PROJECT REPORT

Project ID:	P019
Date:	28-1-26
Project Name:	Comprehensive setup for temperature, humidity, and gas sensor fusion for Indoor Air Quality (IAQ)
Student Name:	Goli Sathwika
Roll No:	122201005
Mentor Name:	Dr. Shaikshavali Chitraganti

1. Project Overview

The main aim of this project is to develop a comprehensive Indoor Air Quality (IAQ) monitoring system that integrates multiple environmental sensors to provide real-time assessment of air quality in indoor spaces. The system will measure critical parameters including carbon dioxide levels, volatile organic compounds, particulate matter, temperature, humidity, and harmful gases to ensure healthy indoor environments.

Indoor air quality has become increasingly important as people spend approximately 90% of their time indoors. Poor IAQ can lead to various health issues including respiratory problems, allergies, reduced cognitive function, and long-term chronic diseases. This monitoring system will enable proactive identification and mitigation of air quality issues in residential, commercial, and industrial settings.

2. Indoor Air Pollutants and Health Impacts

Indoor air quality is influenced by the concentration of key pollutants identified by the Central Pollution Control Board (CPCB), India, as major health risk factors. Carbon dioxide **(CO_2)** serves as an indicator of ventilation adequacy, with levels above 1000 ppm causing discomfort and reduced cognitive performance. Volatile Organic Compounds **(VOCs)** emitted from household and building materials can cause irritation and long-term health risks. Particulate matter **(PM1.0, PM2.5, and PM10)** from indoor activities and outdoor infiltration can penetrate deep into the respiratory system, leading to respiratory and cardiovascular diseases.
Carbon monoxide **(CO)** is a toxic by-product of incomplete combustion that impairs oxygen transport and can be fatal at high concentrations. Nitrogen dioxide **(NO_2)** and ammonia **(NH_3)** from combustion and chemical sources irritate the airways and reduce lung function with prolonged exposure.

3. Air Quality Index (AQI) Calculation

The formula to calculate AQI is the same as per the Indian CPCB and US-EPA. The AQI is calculated using the equations separately for parameters. For example, if we wish to calculate AQI on the basis of four parameters, use the equation four times, and the worst sub-index communicates the AQI. A subindex is a linear function (two different yet related notions) of the concentration of pollutants.
The US EPA AQI uses a piecewise linear function:

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{HI} - BP_{Lo}}(C_p - BP_{Lo}) + I_{Lo}.$$

Where:

C_p = measured pollutant concentration,
BP_{HI} = concentration breakpoint i.e. greater than or equal to Cp
BP_{Lo} = concentration breakpoint i.e. less than or equal to Cp
I_{HI} = AQI value corresponding to BP_{Hi}
I_{Lo} = AQI value corresponding to BP_{Lo}

India uses a sub-index calculation for each pollutant:

The Indian AQI range differs from that of US-EPA. To calculate AQI, a minimum of three parameters should be taken out of which one must be either PM10 or PM2.5 and identified that India AQI requires minimum three parameters (one must be PM10 or PM2.5) with 16-hour data windows for sub-index calculation. The overall AQI is determined as the maximum value among all calculated sub-indices.

The overall AQI is the maximum of all sub-indices.

4. Sensor Selection and Technical Specifications

After evaluation of available sensor technologies, the following sensors have been selected for the IAQ monitoring system based on accuracy, reliability, power consumption, and cost-effectiveness:

Component	**Pollutants Detected**	**Interface**
SCD40	CO_2, Temperature, Humidity	I2C
ENS160	VOC (Volatile Organic Compounds)	I2C
PMS5003	Particulate Matter (PM1.0, PM2.5, PM10)	UART
MiCS-6814	CO, NO_2, NH_3	Analog

Arduino MKR NB 1500	NB-IoT/LTE-M Connectivity	-

5. Sensor Selection

The SCD40 sensor was chosen for its NDIR-based CO_2 measurement providing accuracy of ±50 ppm with built-in temperature and humidity compensation.

The ENS160 provides multi-gas detection for VOC quantification.

The PMS5003 uses laser scattering technology for precise particulate matter measurement across three size ranges.

The MiCS-6814 integrates three independent sensors for hazardous gas detection.

While the Arduino MKR NB 1500 enables cellular connectivity for remote deployment scenarios.

6. Monthly Progress Summary

Pollutant Research: Studied seven major indoor air pollutants (CO_2, VOCs, PM1.0/2.5/10, CO, NO_2, NH_3) and their health impacts using WHO guidelines, EPA standards, and CPCB reports.

AQI Methodology: Researched US EPA and India AQI calculation formulas. Documented the piecewise linear interpolation method

Sensor Selection: Evaluated and compared multiple sensors across each category. Finalized four sensors based on accuracy, cost, and interface compatibility: SCD40 (CO_2/Temp/Humidity), ENS160 (VOC), PMS5003 (PM), and MiCS-6814 (CO/NO_2/NH_3).

Hardware Planning: Mapped sensor interfaces (2× I2C, 1× UART, 1× Analog) and selected Arduino MKR NB 1500 for cellular connectivity. Created List of materials

Feedback from the mentor/s:

Signature of the mentor/s with Date: 29/01/26

OELP - Monthly Progress Report

Project ID:	P019
Date:	26-2-2025
Project Title:	Comprehensive setup for temperature, humidity, and gas sensor fusion for Indoor Air Quality (IAQ)
Student name	Goli Sathwika (122201005)
Mentor:	Dr.Shaikshavali Chitraganti

Work done in the current month:

1. Project Overview & Hardware Setup

Built a real-time **Indoor Air Quality (IAQ) monitor** on an ESP32 that measures 7 pollutants across 4 sensors, computes a composite AQI per different standards for indoor air quality, and streams data to the cloud. Sensors are connected over two daisy-chained I²C buses and one UART interface which is verified for correct addressing and no bus contention:

Sensor	**Measures**	**Bus**	**Pins (ESP32)**
SCD4x	CO_2, Temperature, Humidity	I²C-1	SDA = GPIO 2, SCL = GPIO 1
ENS160	TVOC, eCO_2	I²C-1	SDA = GPIO 2, SCL = GPIO 1
MiCS-4514	CO, NO_2, NH_3	I²C-2	SDA = GPIO 3, SCL = GPIO 4
PMS5003	PM1.0, PM2.5, PM10	UART	RX = GPIO 18, TX = GPIO 17

2. Sensor Testing, Signal Processing & AQI Computation

Each sensor was validated individually before integration to ensure reliable performance.

- **SCD4x:** CO_2 increased from ~430 ppm (ambient) to ~800 ppm in a closed room with no ventilation, confirming response. Temperature and humidity were verified against a reference. ENS160 eCO_2 is cross-checked each cycle, and >500 ppm deviation triggers an alert.
- **ENS160:** TVOC spiked immediately on exposure to perfume and isopropyl alcohol, confirming VOC detection. Operates on the same I²C bus as SCD4x without issues.
- **MiCS-4514:** Under normal indoor conditions, the sensor readings stayed steady, and when exposed , it reacted noticeably, showing that it is able to detect gas presence.
- **PMS5003:** PM2.5 and PM10 increased near burning incense. A 30-second startup delay allows fan and laser stabilization before measurements.
- Since raw readings fluctuated significantly, a 6-sample rolling average (one reading every 5 seconds, 30-second window) is used to smooth outputs before AQI calculation. Gas concentrations are converted to standard units at 25°C wherever required

- For AQI, a sub-index is calculated for each pollutant using linear interpolation over NAQI breakpoints. The final AQI is the highest sub-index, and the dominant pollutant is identified and logged each cycle.

Breakpoint Table:

AQI	**Category**	**CO2(ppm)**	**TVOC(ppb)**	**CO(ppm)**	**NO_2 (µg/m³)**	**PM2.5 (µg/m³)**	**PM10 (µg/m³)**	**NH_3 (µg/m³)**
0-100	Good	400–999	0–249	0–8.7	0–99	0–50	0–75	0–400
100-200	Moderate	1000–1499	250–349	8.8–10	100–299	51–90	76–125	401–800
201-300	Polluted	1500-2499	350-399	10.1-15	300-349	91-140	126-200	801-1200
301-400	Very Polluted	2500–3999	400–499	15.1–30	350–499	141–200	201–300	1201–1800
401-500	Severely Polluted	≥4000	≥500	>30	≥500	>200	>300	>1800

3. Cloud Integration using ThingSpeak via Wi-Fi

- Data pushed to ThingSpeak every 20 s over HTTP GET using the onboard ESP32 Wi-Fi module.
- 8 fields uploaded per cycle: CO_2 (ppm), TVOC (ppb), CO (mg/m³), NH_3 (µg/m³), PM2.5 (µg/m³), PM10 (µg/m³), Temperature (°C), and Final AQI.
- Live time-series graphs monitored on the ThingSpeak dashboard with CO_2 build-up in a closed room and a sharp PM2.5 spike from cooking were clearly captured in the logged data.

Next, we plan to design a custom PCB for a compact and reliable setup, develop a web dashboard for real-time data monitoring, and integrate a SIM-based module so the device can work without Wi-Fi and be deployed in remote locations.

Feedback from the mentor/s:

Signature of the mentor with Date:

C. [illegible]
26/02/26.

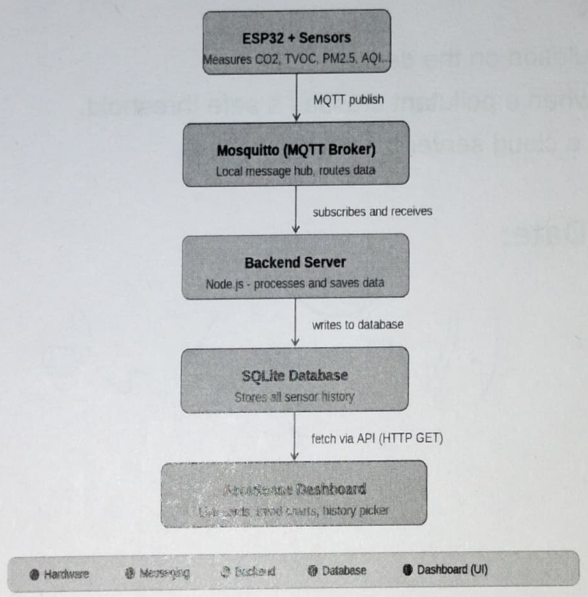
OELP - Monthly Progress Report

Project ID:	P019
Date:	30-3-2026
Project Title:	Comprehensive setup for temperature, humidity, and gas sensor fusion for Indoor Air Quality (IAQ)
Student name:	Goli Sathwika (122201005)
Mentor:	Dr.Shaikshavali Chitraganti

Work done in the current month:

1. System Architecture → How Data Flows

The diagram below shows how sensor data travels from the physical device all the way to the web dashboard.



2. Backend Server (Node.js + Express.js + SQLite)

The backend is a lightweight API server. It receives sensor data from the dashboard, stores it permanently in a local database, and serves it back whenever the dashboard needs to display readings or history. It acts as a reliable middleman and historian ,catching streaming data, filing it neatly with an accurate server-side timestamp, and allowing the frontend to fetch either the latest snapshot or a full day's worth of data on demand.

It is made up of three files:

- **index.js** – The main server. Sets up Express on port 3001, enables CORS (so the browser-based dashboard can talk to it), and defines the three API endpoints.
- **db.js** – Handles the database. Creates the local iaq_data.db file and the sensor_data table (with columns for AQI, CO_2, TVOC, PM2.5, PM10, temperature, humidity, and more) if they don't already exist.

3. The Three API Endpoints

- Developed a **POST /api/history** endpoint to receive sensor readings from the dashboard, assign server-side timestamps, and store the data in the database.
- Created a **GET /api/history** endpoint to retrieve the latest 100 records for real-time visualization in the dashboard.
- Built a **GET /api/history/:date** endpoint to fetch historical data for a selected date by converting it to IST and returning the corresponding records.

4. Dashboard (React + Vite):

- Integrated MQTT communication to receive live readings from the ESP32 via the Mosquitto broker and forward them to the backend using the **POST /api/history** endpoint for storage.
- Implemented live metric cards to display current values of CO_2, TVOC, PM2.5, PM10, CO, NO_2, temperature, and humidity, along with a colour-coded overall AQI status indicator.
- Built a historical trends module with a date picker to fetch and visualize past data, including interactive charts and toggle options for different pollutants.

Plan for Next Month:

- Fine-tune the AQI calculation on the device.
- Add dashboard alerts when a pollutant crosses a safe threshold.
- Deploy the backend to a cloud server for remote access.

Mentor Signature & Date:

30/03/26

OELP - Monthly Progress Report

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Student name	Goli Sathwika (122201005)
Mentor:	Dr.Shaikshavali Chitraganti

Work Done In current Month:

- **Deployed the dashboard frontend on Render:** The React-based dashboard is now live and publicly accessible via Render. Users can monitor real-time air quality data from any device and browser without needing a local setup. .

- **Migrated to full cloud infrastructure using Supabase and Render:** The local SQLite database has been replaced with Supabase PostgreSQL for permanent cloud storage, real-time subscriptions, and historical analytics. The Node.js bridge script is deployed on Render, running continuously in the background to listen to MQTT topics and push sensor readings to the database — eliminating all dependency on a local machine and ensuring uninterrupted 24/7 data logging.

- **Implemented weekly and 15-day metric averaging** with bar graph visualization: Historical sensor data (temperature, humidity, and gas concentrations) is now aggregated over weekly and 15-day intervals. These computed averages are displayed as interactive bar graphs on the dashboard, enabling users to identify trends, seasonal patterns, and anomalies over time.

- **Designed and 3D printed a custom hardware enclosure**: A protective case was designed and fabricated using 3D printing to house all electronic components, including the sensors, microcontroller, and wiring. The enclosure ensures physical protection and a clean, deployable form factor suitable for real indoor environments.

Mentor Signature & Date:

C. Thal 30/04/2026.